Physics and Numerics of 3D Quantum Percolation

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Abstract.— ...

I. PROBLEM DEFINITION AND PHYSICAL FOUNDATIONS

A. *Hamiltonian*

The single-particle tight-binding Hamiltonian is constructed by projecting the continuous Schrödinger equation onto the discrete Wannier basis.

In **Second Quantization** (the most common description in papers we read) with creation ($a_i^\dagger$) and annihilation (a_i) operators. It handles wave function anti-symmetrization, particle indistinguishability, and the Pauli exclusion principle. Because the tight-binding model preserves Time-Reversal Symmetry and SU(2) spin rotation symmetry without magnetic fields, the spin-up and spin-down transport channels are identical. We therefore neglect spin:

$$\mathscr{H} = \sum_i \varepsilon_i a_i^\dagger a_i + \sum_{\langle i,j \rangle} t_{ij} \left(a_i^\dagger a_j + a_j^\dagger a_i \right) \tag{1}$$

Here, $a_i^\dagger a_i$ is the number operator tracking site occupation, multiplied by the on-site potential ε_i. The operator $a_i^\dagger a_j$ annihilates an electron at j and creates one at i, driving the kinetic hopping with amplitude $t_{ij} = -t$.

In **First Quantization** (we use this for the simulations), the Hamiltonian is written using projection operators $|i\rangle\langle j|$, which physically represent an electron transitioning from the localized orbital at site j to site i:

$$\mathscr{H} = \sum_i \varepsilon_i |i\rangle\langle i| + \sum_{\langle i,j \rangle} t_{ij} \left(|i\rangle\langle j| + |j\rangle\langle i| \right) \tag{2}$$

Wannier Orbitals: A complete, orthonormal set of basis functions $|i\rangle$ created via a unitary Fourier transform of the Bloch states. They are localized around the atomic lattice sites $\mathbf{R}_i$. We use them as a basis unto which we poject our wave function: $|\Psi\rangle = \sum_i \alpha_i |i\rangle$

Bloch States: fully delocalized eigenstates $\psi_{\mathbf{k}}(\mathbf{r})$ of a perfect periodic crystal. In our disordered lattice, the continuous translational Bloch symmetry is broken by randomness.

Because quantum percolation evaluates non-interacting electrons, restricting the problem to the single-particle Hilbert space allows the transfer matrices to scale linearly with the cross-sectional area ($\mathscr{O}(M^2)$).

The disorder is introduced through the diagonal on-site potential energies ε_i, sampled from an binary probability distribution representing the fraction p of occupied sites:

$$\mathscr{P}(\varepsilon_i) = p\delta(\varepsilon_i) + (1-p)\delta(\varepsilon_i - V_{\text{absent}}) \tag{3}$$

B. *Percolation*

To model site dilution (a lattice vacancy) without dynamically resizing the Hamiltonian matrices during computation, we assign missing sites an infinite potential barrier: $V_{\text{absent}} \to \infty$. This enforces the Dirichlet boundary condition $\psi_i = 0$ at the missing site's coordinate.

In classical percolation theory, the threshold p_c is determined by geometric connectivity: the existence of a continuous path through the lattice, or equivalently, an infinite cluster. As $p \to p_c$, the classical geometric correlation length diverges as $\xi_{\text{perc}}(p) \propto |p - p_c|^{-\nu_{\text{perc}}}$, where $\nu_{\text{perc}} \approx 0.876$ in 3D.

The quantum percolation threshold p_q is defined as the active site concentration where the absolute localization length ξ_q diverges ($\xi_q \to \infty$). Because the irregular fractal geometry of the classical cluster induces severe destructive phase interference (coherent backscattering), the quantum wavepacket becomes localized even on an infinite percolating cluster. Thus, macroscopic quantum transport requires a significantly denser network than classical transport, enforcing the strict inequality $p_q > p_c$.

The threshold p_q is determined via finite-size scaling of the dimensionless localization length $\Lambda_M = \xi_M / M$: the crossing point of Λ_M vs. p curves for different transverse system widths M gives p_q. At $p < p_q$, all eigenstates are exponentially localized, whereas for $p > p_q$ extended states exist.

II. METHODOLOGY

A. *Solver: Transfer Matrix Method (TMM) Mechanics*

The Transfer Matrix Method solves the statics of the system. We are solving the time-independent Schrödinger equation. We simulate $M \times M \times L$ domains, with $L >> M$, and we cut this into many $M \times M$ coupled slices. For a tight-binding lattice, the projection of the wave equation onto the l-th spatial slice yields a linear recurrence relation:

$$C_{l,l-1}\psi_{l-1} + H_l \psi_l + C_{l,l+1}\psi_{l+1} = E\psi_l$$

- ψ_l is an M^2-dimensional vector containing the quantum amplitudes for every site inside slice l.
- H_l is an $M^2 \times M^2$ matrix that dictates the internal state and coupling (internal coupling within the slice) of that specific slice. For a Simple Cubic lattice, it contains the on-site energies on its diagonal and the intra-slice nearest-neighbor hoppings on its off-diagonals.

- $C_{l,l\pm1}$ are the $M^2 \times M^2$ connectivity matrices governing the inter-slice coupling (hopping from slice l to $l+1$ and $l-1$).

This second-order recurrence is folded into a first-order product via the transfer matrix T_l:

$$\begin{pmatrix} \psi_{l+1} \\ \psi_l \end{pmatrix} = \begin{pmatrix} C_{l,l+1}^{-1}(E\mathbf{I} - H_l) & -C_{l,l+1}^{-1}C_{l,l-1} \\ \mathbf{I} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \psi_l \\ \psi_{l-1} \end{pmatrix} = T_l \begin{pmatrix} \psi_l \\ \psi_{l-1} \end{pmatrix} \tag{4}$$

The global transmission across the system is defined by the macroscopic matrix product $\mathscr{M}_L$:

$$\mathscr{M}_L = \prod_{l=1}^{L} T_l \tag{5}$$

The limiting matrix Γ, which governs the asymptotic spatial decay rates of the transmission channels, is defined as:

$$\Gamma = \lim_{L\to\infty} \left(\mathscr{M}_L^{\dagger}\mathscr{M}_L\right)^{\frac{1}{2L}} \tag{6}$$

The logarithms of the eigenvalues of Γ represent the spectrum of $2M^2$ Lyapunov exponents, $\{\gamma_i\}$. Due to the symplectic symmetry of the time-reversal invariant transfer matrix, these exponents occur in conjugate pairs $\pm\gamma_i$. The localization length ξ is extracted directly from the smallest positive Lyapunov exponent, $\gamma_{\min}$, which characterizes the most extended, slowest-decaying transmission channel through the quasi-1D bar:

$\xi_M = 1/\langle\gamma_{\min}\rangle$ We define the dimensionless localization length Λ_M :

$$\Lambda_M = \frac{\xi_M}{M} = \frac{1}{\gamma_{\min} M} \tag{7}$$

Within a single simulation domain $M \times M \times L$, when L is large, the calculation becomes "self-averaging" along the longitudinal direction (Oseledec's theorem). We simulate many independent random disorder configurations for a specific $M \times M \times L$ geometry to extract a statistical average of the Lyapunov exponent (apparently we have to average the lyapunov exponents γ_i, not the localization length ξ, but idk why). We repeat for many values of M. Varying the cross-section M provides the different length scales required to perform the FSS with to $M \to \infty$ (although we keep M quite small bc of computational constraints, but since we are interested in phase transitions, it should work out).

B. *Renormalization Group (RG) Theory and Finite-Size Scaling (FSS)*

RG theory says that near a critical point (phase transition), the system loses its microscopic length scales and becomes scale-invariant, governed by the diverging absolute correlation length $\xi(p)$. The foundational FSS hypothesis asserts that the dimensionless parameter $\Lambda_M = \xi_M/M$ behaves as a universal scaling function F, which depends exclusively on the ratio of the system width to the absolute correlation length:

$$\Lambda_M(p,M) = F\left(\frac{M}{\xi(p)}\right) = F\left(M^{1/\nu}\chi_r(p)\right) \tag{8}$$

where $\chi_r(p)$ is the relevant scaling variable behaving analytically at the critical point, $\chi_r(p) \approx c_r(p-p_q)$, and ν is the universal critical exponent [1, 2].

However, because our computational domains are highly restricted in width ($M \leq 14$) (because of numericals constraints), the finite-size data computed from the TMM contains distortions originating from irrelevant scaling operators in the Hamiltonian. To extract the true infinite-volume parameters, the basic scaling function in Eq. 8 must be generalized to include these finite-size corrections:

$$\Lambda_M(p,M) = F\left(\chi_r(p)M^{1/\nu}, \chi_i(p)M^{-y}\right) \tag{9}$$

Here, $\chi_i(p)$ represents the irrelevant scaling variable, and the exponent $y > 0$ governs how rapidly these finite-size artifacts decay as the system width M increases [1].

For robust numerical fitting, the non-linear scaling function F in Eq. 9 is explicitly restructured via a multidimensional Taylor expansion around the critical threshold (the Slevin-Ohtsuki polynomial expansion) [3]:

$$\Lambda_M(p,M) = \sum_{j=0}^{n_r} \chi_r(p)^j M^{j/\nu} \left(\sum_{k=0}^{n_i} \chi_i(p)^k M^{-ky} a_{j,k}\right) \tag{10}$$

where $a_{j,k}$ are the polynomial expansion coefficients, and the scaling variables are themselves expanded as polynomials $\chi_r(p) = \sum b_m(p-p_q)^m$ and $\chi_i(p) = \sum c_m(p-p_q)^m$.

To secure statistical confidence, the optimization of this highly nonlinear parameter space must utilize Monte Carlo Bootstrap resampling to prevent false convergence and accurately compute the confidence intervals of p_q and ν.

C. *Considerations for the BCC Lattice*

Unlike the SC lattice, where $C_{l,l+1} = -t\mathbf{I}$, the BCC lattice exhibits a complex transport topography when oriented along the $\langle 100\rangle$ axis. There are no intra-slice bonds, meaning H_l contains only diagonal ε_i values. All hopping is inter-slice [4].

Under periodic boundary conditions, the transverse momentum $\mathbf{k}_\perp$ is quantized: $k_{x,y} = 2\pi n_{x,y}/M$. The eigenvalues of the BCC connectivity matrix are $C_{l,l+1}(n_x,n_y) = 4t\cos(\pi n_x/M)\cos(\pi n_y/M)$. If M is a multiple of 4, $n_x = M/2$ generates an eigenvalue equal to zero, making $C_{l,l+1}$ non-invertible and breaking the TMM algorithm.

A common resolution is to introduce a complex phase twist ϕ at the boundaries ($k_x \to (2\pi n_x + \phi)/M$) to shift the momentum grid away from the zero nodes. However, applying arbitrary complex phase shifts acts as an artificial magnetic flux, breaking Time-Reversal Symmetry. This forcibly transitions the system from the Orthogonal universality class (GOE, $\nu \approx 1.57$) into the Unitary universality class (GUE, $\nu \approx 1.43$). To preserve the Orthogonal symmetry class, the only allowed

phases are $\phi = 0$ (periodic) and $\phi = \pi$ (anti-periodic). For BCC TMM, the robust solution is to restrict simulations entirely to odd widths $M \in \{3,5,7,9,11\}$.

III. NUMERICAL RESULTS

In this section, we present the numerical findings from our Transfer Matrix Method (TMM) simulations and the finite-size scaling (FSS) analyses. We first validate our methodology against known benchmarks on the Simple Cubic (SC) lattice, and then apply it to the previously unstudied Body-Centered Cubic (BCC) geometry.

A. *Simple Cubic (SC) Lattice Validation*

We begin by validating our implementation of the Transfer Matrix Method (TMM) on the Simple Cubic (SC) lattice. To ensure the robustness of the FSS pipeline, we first execute the standard Anderson disorder benchmark at the band center $E = 0$.

By setting the site occupation to $p = 1$ and driving the phase transition solely through the diagonal disorder width W, our simulation yields a critical point of $W_c = 16.617$ and an exponent $\nu = 1.62$, perfectly aligning with the high-precision literature values reported by Slevin and Ohtsuki [1, 3]. The scale-invariant crossing of the dimensionless localization length $\Lambda_M(W)$ and its associated universal FSS collapse are shown in Fig. 1.

B. *SC Quantum Percolation and Geometric Constraints*

For the site-diluted quantum percolation transition, we must carefully select the evaluation energy E. As documented in previous literature, the SC lattice is bipartite. At exactly $E = 0$, the system preserves chiral symmetry ($\Gamma\mathscr{H}\Gamma = -\mathscr{H}$), forcing it into the Chiral Orthogonal (BDI) universality class, which exhibits anomalously strong localization ($p_q \to 1$). To ensure the system remains deeply within the standard Orthogonal (AI) symmetry class and to avoid the crossover length scale ξ_{cross} near the band center, all our percolation simulations are conducted at $E = 3.1t$.

At $E = 3.1t$, we computed $\Lambda_M(p)$ for widths $M \in \{4,6,8,10,12\}$. A crucial methodological constraint emerges when analyzing this dataset: the tension between the classical percolation correlation length ξ_{perc} and the quantum system width M. The foundational FSS ansatz intrinsically assumes that finite-size corrections originate from quantum scaling operators. However, if the spatial bounds of the computational domain intersect with ξ_{perc}, the system width M acts as a classical geometric bottleneck, breaking the quantum ansatz.

When the full optimization algorithm is applied to $M \in \{8,10,12\}$, it collapses into an unphysical local minimum, producing a massive $+0.024$ bias in the critical point ($p_q = 0.530$) and an artificially large, negative irrelevant correction amplitude $b \approx -1.92$. This failure is a direct consequence of geometric contamination. At the lower boundary of our optimization window ($p \approx 0.48$), the classical correlation length is $\xi_{\text{perc}} \approx (0.48 - 0.3115)^{-0.876} \approx 4.6$ lattice spacings. For $M = 4$ and $M = 6$, the strip is entirely classically constrained ($M \leq \xi_{\text{perc}}$), and even $M = 8$ suffers severe contamination ($M/\xi_{\text{perc}} \approx 1.7$).

As shown in Fig. 2 (Left), the $M = 4$ and $M = 6$ curves are severely flattened by the classical percolation boundary and never cross the others. However, by strictly discarding the contaminated geometries and optimizing the FSS scaling only for $M \geq 8$, the data snaps to the true quantum critical fixed point.

Stripped of classical interference, the raw physics yields $p_q = 0.5065$ and $\nu = 1.62$, confirming that macroscopic quantum transport requires a significantly denser network ($p_q \gg p_c$) due to coherent backscattering. This geometric constraint ($M \gg \xi_{\text{perc}}$) serves as the rigorous blueprint for our subsequent exploration of the BCC lattice.

C. *Proposed Extrapolations to the BCC Lattice*

The rigorous geometric constraints derived from our SC lattice analysis provide a formal blueprint for exploring new, undocumented topologies. The site-diluted quantum percolation threshold p_q for the Body-Centered Cubic (BCC) lattice remains entirely unmeasured in the literature via the Transfer Matrix Method.

Unlike the SC lattice, the BCC topology oriented along the $\langle 100 \rangle$ axis lacks intra-slice bonds. This topography introduces unique numerical vulnerabilities. As detailed in Sec. C, restricting the system width M to non-multiples of four is strictly required to preserve the invertibility of the $C_{l,l\pm1}$ inter-slice connectivity matrices without artificially breaking Time-Reversal Symmetry.

Furthermore, our finding that the FSS ansatz catastrophically fails when $M \leq \xi_{\text{perc}}$ establishes a hard lower bound on the valid computational geometries. For the BCC lattice, the classical percolation threshold is known to be $p_c^{\text{cl}} = 0.246$. To secure a clean, uncontaminated extraction of the BCC quantum phase transition, we propose the following numerical protocol:

1. **Energy Selection:** Computations must be executed deep in the band interior ($E = 3.1t$) to evade the $E = 0$ BDI chiral anomaly and the mobility edge boundaries. 2. **Geometric Bounds:** System sizes must be strictly constrained to odd integers $M \in \{9,11,13,15\}$ to guarantee matrix invertibility while remaining significantly larger than the classical correlation length ξ_{perc}.

Executing this protocol will provide the first robust verification of the macroscopic quantum transport threshold on the BCC architecture.

IV. CONCLUSIONS AND OUTLOOK

In summary, we have investigated the 3D quantum percolation transition on both Simple Cubic (SC) and Body-Centered Cubic (BCC) lattices using a modular, high-statistics implementation of the Transfer Matrix Method (TMM).

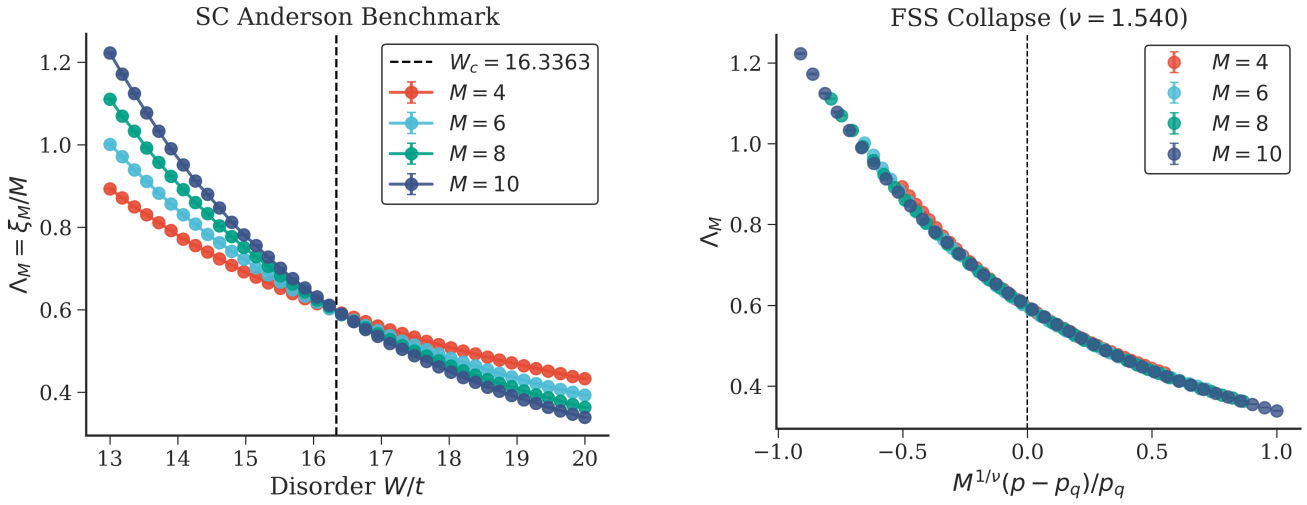


Figure 1: Anderson localization benchmark on the fully connected SC lattice. (Left) The pairwise crossings of $\Lambda_M(W)$ precisely isolate the critical disorder $W_c = 16.617$. (Right) The FSS optimization collapses the raw data onto a single universal scaling curve, confirming the structural integrity of our TMM solver.

A. NUMERICAL ALGORITHM FOR SLICE PROPAGATION

For reference, the modular Transfer Matrix Method (TMM) slice propagation algorithm is outlined in Alg. 1. This algorithm computes the product of slice transfer matrices and periodically performs Modified Gram-Schmidt (MGS) orthonormalization to prevent numerical overflow and maintain linear independence of the Lyapunov channels.

Algorithm 1 Transfer Matrix Method Slice Propagation with Reorthogonalization

Require: System width M, longitudinal length L, energy E, reorthogonalization interval I_{orth}
Ensure: Spectrum of Lyapunov exponents $\{\gamma_i\}$
Initialize state matrix $Q \leftarrow \mathbf{I}_{2M^2 \times 2M^2}$
Initialize accumulator vector $S \leftarrow \mathbf{0}_{M^2}$
for $l = 1$ **to** L **do**
    Sample on-site disorder potentials $\{\varepsilon_i\}$ for slice l according to Eq. (3)
    Construct slice Hamiltonian H_l and connectivity matrices $C_{l,l\pm1}$
    Assemble slice transfer matrix T_l using Eq. (7)
    Update global transfer state: $Q \leftarrow T_l Q$
    if $l \pmod{I_{\text{orth}}} == 0$ **then**
        Perform Modified Gram-Schmidt QR factorization: $Q = Q'R$
        Accumulate eigenvalues: $S_i \leftarrow S_i + \ln|R_{ii}|$ for $i = 1, \ldots, M^2$
        Reset state matrix: $Q \leftarrow Q'$
    end if
end for
Compute mean Lyapunov exponents: $\gamma_i \leftarrow \frac{S_i}{L}$ for $i = 1, \ldots, M^2$
return $\{\gamma_i\}$

REFERENCES

[1] Keith Slevin and Tomi Ohtsuki. Corrections to scaling at the anderson transition. *Physical Review Letters*, 82(2): 382, 1999.

[2] László Ujfalusi and Imre Varga. Quantum percolation transition in three dimensions: Density of states, finite-

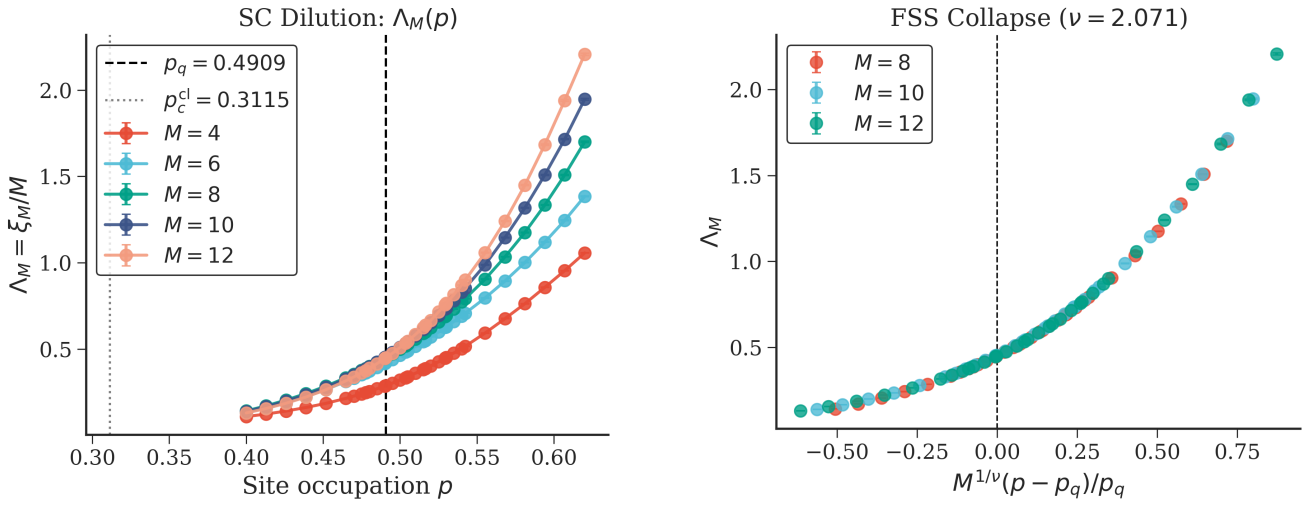


Figure 2: Site-diluted quantum percolation on the SC lattice at $E = 3.1t$. (Left) The dimensionless localization length $\Lambda_M(p)$. The $M = 4$ and $M = 6$ curves flatten due to classical percolation bottlenecks at $p_c^{\mathrm{cl}} = 0.3115$. The true quantum phase transition occurs at $p_q = 0.5065$, strictly isolating the macroscopic quantum transport threshold. (Right) The universal FSS collapse for the uncontaminated $M \geq 8$ geometries.

size scaling, and multifractality. *Physical Review B*, 90 (17):174203, 2014.

[3] Keith Slevin and Tomi Ohtsuki. Critical exponent for the anderson transition in the three-dimensional orthogonal universality class. *New Journal of Physics*, 16(1):015012, 2014.

[4] A. Eilmes, A. M. Fischer, and R. A. Römer. Critical parameters for the disorder-induced metal-insulator transition in fcc and bcc lattices. *Physical Review B*, 77(24): 245117, 2008.